



Project Title :VLAN and Inter-VLAN Routing Using FortiGate

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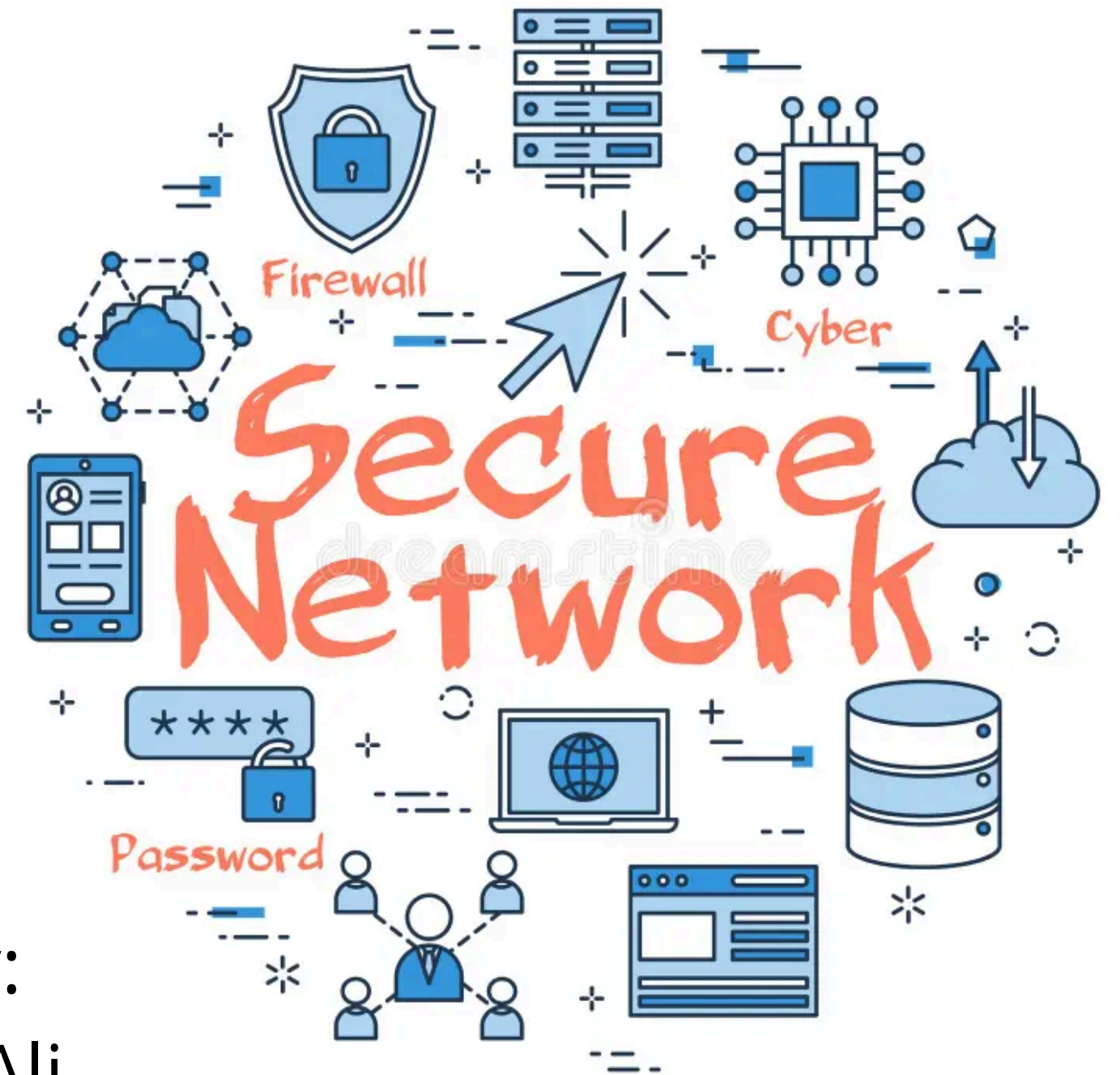


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PROJECT OVERVIEW

- Implement VLANs on a switch
- Integrate FortiGate for routing
- Apply firewall policies
- Test communication between VLANs

FORTIGATE FIREWALL

Firewall Evolution

Transparent Firewall

- This firewall does not have an IP address
- Work as a layer 2 switch
- No NAT
- No routing
- Inspection of traffic up to layer 2

What is a Firewall

- A firewall is a network security device that monitors and filters incoming and outgoing network traffic based on an organization's previously established security policies.
- A firewall's main purpose is to allow non-threatening traffic in and to keep dangerous traffic out.

First & Second Generation (Traditional Firewall).

- Work at OSI Layer 3 & 4
- NAT
- Routing
- Traffic Inspection
- VPN
- No IPS
- Do not have application awareness

UTM (Unified Threat Management)

Benefits

- Work at OSI Layer 7
- All-in-one network security platforms.
- Firewall - VPN
- IPS-IDS

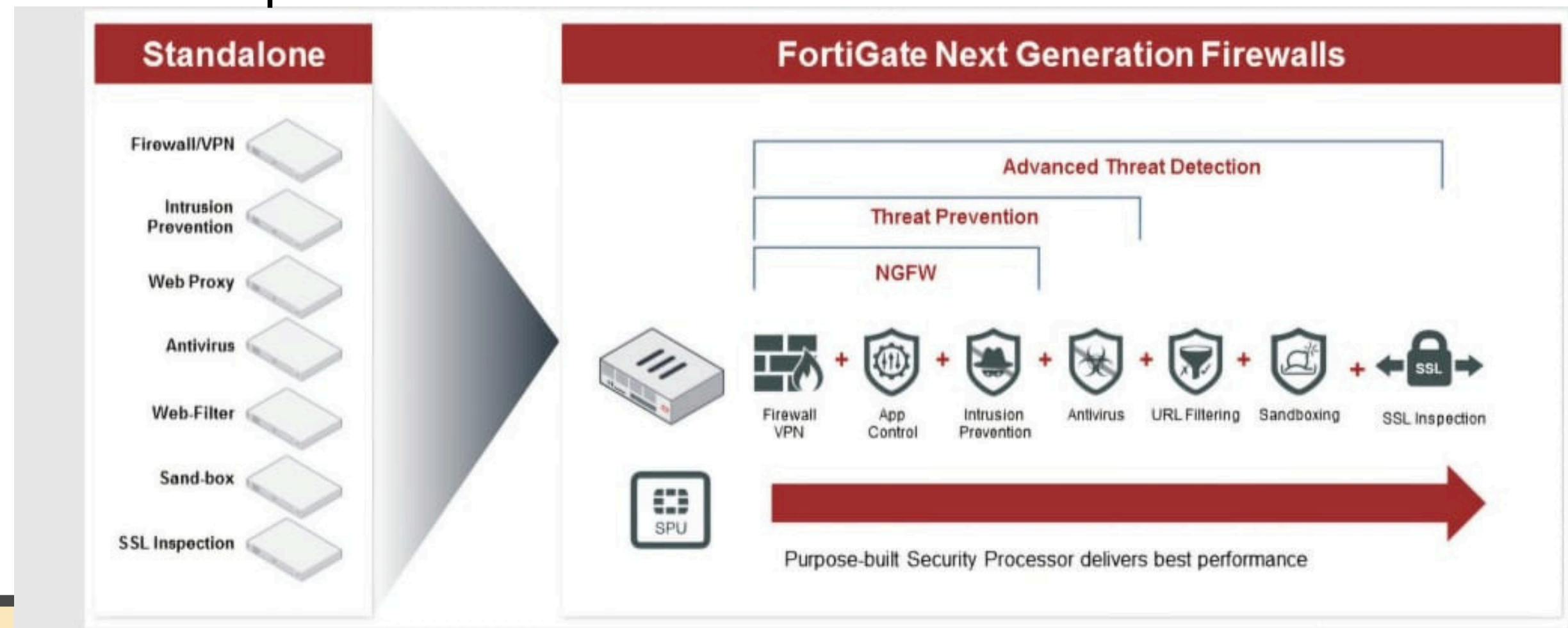
Antivirus-Antispyware

- Web Filtering-Mail Filtering

NGFW (Next Generation Firewall)

It is very similar to UTM, but with limited feature

That means NGFWs may include IPS, antivirus and malware prevention, application control, deep packet inspection and stateful firewalls, encryption, compression, QoS and other capabilities



VLANs

- What is VLAN?

A VLAN (Virtual LAN) allows devices on the same physical network to be grouped into separate virtual networks.

👉 Improves security, performance, and traffic management.

- Why do we need VLANs?

We use VLANs to separate departments within a company:

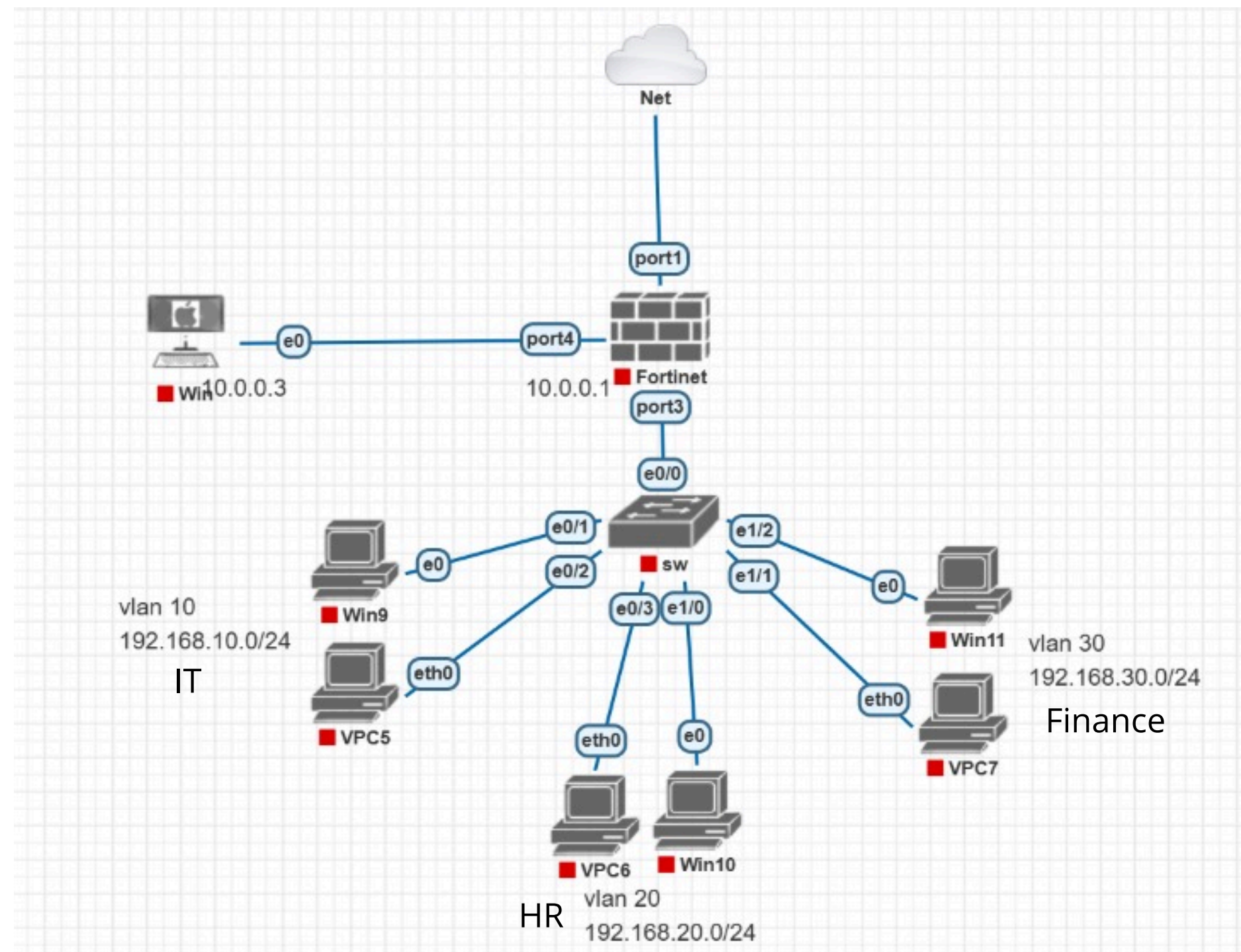
- VLAN 10 → IT Department
- VLAN 20 → HR
- VLAN 30 → Guests

Each VLAN has its own IP range and is isolated from the others.

VLANs

- Benefits of using VLANs
- What is Inter-VLAN Routing

NETWORK TOPOLOGY



Three VLANs are connected to a FortiGate firewall through a Layer 2 switch

VLAN CONFIGURATION ON SWITCH

1. Create vlans on switch and Configured Trunk port

```
Switch>enable
Switch#sh
Switch#show vlan br
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Et1/3, Et2/0, Et2/1, Et2/2 Et2/3, Et3/0, Et3/1, Et3/2 Et3/3, Et4/0, Et4/1, Et4/2 Et4/3, Et5/0, Et5/1, Et5/2 Et5/3, Et6/0, Et6/1, Et6/2 Et6/3, Et7/0, Et7/1, Et7/2 Et7/3, Et8/0, Et8/1, Et8/2 Et8/3, Et9/0, Et9/1, Et9/2 Et9/3
10	VLAN0010	active	Et0/1, Et0/2
20	VLAN0020	active	Et0/3, Et1/0
30	VLAN0030	active	Et1/1, Et1/2
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
Switch#
```

```
Switch#s
Switch#sh
Switch#show in
Switch#show inte
Switch#show interfaces t
Switch#show interfaces tr
Switch#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Et0/0	on	802.1q	trunking	1

```
Port
Et0/0
```

Vlans allowed on trunk
1,10,20,30

```
Port
Et0/0
```

Vlans allowed and active in management domain
1,10,20,30

```
Port
Et0/0
```

Vlans in spanning tree forwarding state and not pruned
1,10,20,30

```
Switch#
```

VLANs were created and ports were assigned to their respective VLANs, as shown in the command output.

FORTIGATE CONFIGURATION STEPS

- Create VLAN Interfaces: Go to Network → Interfaces → Create New (VLAN) Assign VLAN IDs (10, 20, 30)
- Configure IP Addressing: Set IP for each VLAN (e.g., 192.168.10.1/24 for VLAN 10)
- Enable DHCP (Optional): Assign DHCP ranges for each VLAN
- Firewall Policies: Allow traffic between VLANs as needed (e.g., HR → Finance, but block Guest)

FORTIGATE CONFIGURATION STEPS

- Created subinterface for each VLANs on FortiGate on port 3

The screenshot displays a PNETLab topology and the configuration of a FortiGate VM. In the background, a network diagram shows a host with IP 192.168.1.9 connected to a switch (e0) with IP 0.0.0.3. The switch is connected to a VLAN 10 (192.168.10.0/24) and a VPC6 (192.168.20.0/24). The foreground shows the FortiGate VM64-KVM interface. A red banner at the top indicates 'FortiGate time is out of sync.' Below this, a table lists the configured interfaces:

Name	Type	Members	IP/Netmask	Administrative Access	Link Status
port3	Physical Interface		0.0.0.0/0.0.0.0		Up
user (vlan 10)	VLAN		192.168.10.1/255.255.255.0	PING HTTPS SSH RADIUS Accounting	Up
user (vlan 20)	VLAN		192.168.20.1/255.255.255.0	PING HTTPS SSH RADIUS Accounting	Up
user (vlan 30)	VLAN		192.168.30.1/255.255.255.0	PING HTTPS SSH RADIUS Accounting	Up

The bottom of the screen shows the Windows taskbar with the time 5:32 PM on 4/4/2025. A 'Physical' timer at the bottom right shows 01 : 00 : 00.

FortiGate policy

What is a Firewall Policy?

A set of rules used to control traffic flow between different networks or VLANs based on criteria like source, destination, service, and action.

Purpose in Our Project:

- Control communication between VLANs
- Restrict unauthorized access
- Allow only specific traffic (e.g., HTTP, DNS)

Enable internet access for certain VLANs

The screenshot shows the 'New Policy' configuration page in the FortiGate web interface. The page is divided into several sections:

- Policy Fields:** Includes fields for Name, Incoming Interface, Outgoing Interface, Source, Destination, Schedule (set to 'always'), Service, and Action (with buttons for ACCEPT, DENY, and IPsec).
- Firewall/Network Options:** Includes NAT (enabled), IP Pool Configuration (Use Outgoing Interface Address), Preserve Source Port (disabled), and Protocol Options (set to 'default').
- Security Profiles:** A list of security profiles with toggle switches: AntiVirus, Web Filter, DNS Filter, Application Control, IPS, File Filter, and SSL Inspection (set to 'no-inspection').

FORTIGATE POLICY

← → ↻ Not secure | 10.0.0.1/ng/firewall/policy/policy/standard ☆ Update

FortiGate time is out of sync.

+ Create New Edit Delete Policy Lookup Search

Export Interface Pair View **By Sequence**

Name	From	To	Source	Destination	Schedule	Service	Action	NAT
TO WAN	user (vlan 10) user (vlan 20)	port1	vlan 10 address vlan 20 address	all	always	ALL	✓ ACCEPT	✓ Enabl
to local network	port1	user (vlan 10) user (vlan 20)	all	all	always	ALL	✓ ACCEPT	✓ Enabl
intervlan routing	user (vlan 10) user (vlan 20) user (vlan 30)	user (vlan 10) user (vlan 20) user (vlan 30)	all	all	always	ALL	✓ ACCEPT	✗ Disab
finance to Wan	user (vlan 30)	port1	vlan 30 address	all	finance	ALL	✓ ACCEPT	✓ Enabl
to finance	port1	user (vlan 30)	all	vlan 30 address	finance	ALL	✓ ACCEPT	✓ Enabl
Implicit Deny	any	any	all	all	always	ALL	✗ DENY	

0 Security Rating Issues 6 Updated: 01:02:42

ENG 1:03 AM

VERIFICATION & TESTING

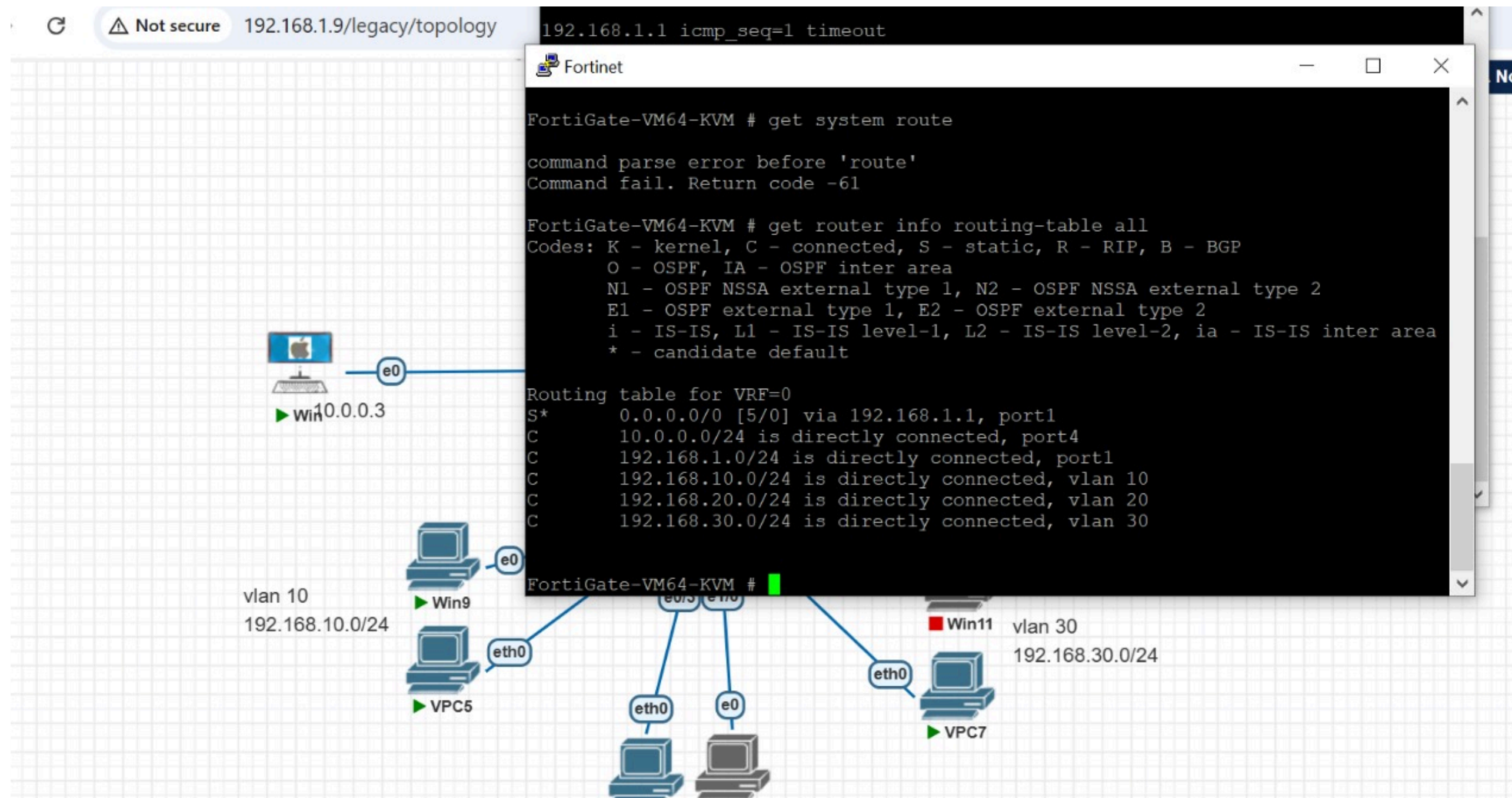
- Ping Test: Verify communication between VLANs

The screenshot displays two VPCS terminal windows and a network diagram. The left terminal window (VPC6) shows the command sequence: `ip dhcp` (resulting in DDORA IP 192.168.20.2/24 GW 192.168.20.1) and `ping 192.168.10.3`. The output of the ping test shows five successful attempts with varying times. The right terminal window (VPC5) shows the command sequence: `ip dhcp` (resulting in DDORA IP 192.168.10.3/24 GW 192.168.10.1) and `ping 192.168.20.2`. The output shows five successful attempts. The network diagram below illustrates the topology: a central switch (SW) with ports e0/2, e0/3, e1/0, and e1/1. It connects to three VLANs: VLAN 10 (192.168.10.0/24) with hosts Win9 and VPC5; VLAN 20 (192.168.20.0/24) with hosts VPC6 and Win10; and VLAN 30 (192.168.30.0/24) with hosts Win11 and VPC7. All hosts are connected to the switch via their eth0 interfaces.

Ping from VLAN 10 to VLAN 20

VERIFICATION & TESTING

- Check Routing Table: get router info routing-table all



VERIFICATION & TESTING

- Logs: Monitor traffic in FortiGate logs

The screenshot displays the FortiGate VM64-KVM web interface. The left sidebar contains a navigation menu with the following items: System, Security Fabric, Log & Report (highlighted), Forward Traffic, Local Traffic, Sniffer Traffic, Events, AntiVirus, Web Filter, SSL, DNS Query, File Filter, Application Control, Intrusion Prevention, Anomaly, and Log Settings. The main content area shows a table of traffic logs. A red banner at the top of the main area indicates "FortiGate time is out of sync." The table has columns for Date/Time, Source, Device, Destination, Result, Action, and Ap. The logs show traffic from 192.168.10.3 and 192.168.20.2 to 192.168.1.1 and 8.8.8.8 (dns.google) via VPCS1, all with a result of "84 B / 84 B" and action of "Accept".

Date/Time	Source	Device	Destination	Result	Action	Ap
7 hours ago	192.168.10.3	VPCS1	192.168.1.1	✓ 84 B / 84 B	Accept	
7 hours ago	192.168.10.3	VPCS1	192.168.1.1	✓ 84 B / 84 B	Accept	
7 hours ago	192.168.10.3	VPCS1	192.168.1.1	✓ 84 B / 84 B	Accept	
7 hours ago	192.168.10.3	VPCS1	192.168.1.1	✓ 84 B / 84 B	Accept	
7 hours ago	192.168.10.3	VPCS1	192.168.1.1	✓ 84 B / 84 B	Accept	
7 hours ago	192.168.20.2	VPCS1	8.8.8.8 (dns.google)	✓ 84 B / 84 B	Accept	
7 hours ago	192.168.20.2	VPCS1	8.8.8.8 (dns.google)	✓ 84 B / 84 B	Accept	
7 hours ago	192.168.20.2	VPCS1	8.8.8.8 (dns.google)	✓ 84 B / 84 B	Accept	
7 hours ago	192.168.20.2	VPCS1	8.8.8.8 (dns.google)	✓ 84 B / 84 B	Accept	
7 hours ago	192.168.20.2	VPCS1	8.8.8.8 (dns.google)	✓ 84 B / 84 B	Accept	

At the bottom of the interface, the FortiGate logo and version v7.0.3 are visible. The status bar at the bottom right shows the language as ENG, the time as 12:38 AM, and the date as 4/10/2025.

CONCLUSION

- VLANs created successfully
- Inter-VLAN routing via FortiGate
- Network segmentation and security improved



THANK YOU

